Live Forensics mit Volatility und Rekall (Windows)

am Beispiel zeus und stuxnet

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Februar 2021

Allgemeines:

Übersicht der Profile:

```
vol.py --info | grep Win
```

```
root@myubu:/tmp/dumpfile# vol.py --info | grep Win
Volatile Systems Volatility Framework 2.3_beta
VistaSP0x64                     - A Profile for Windows Vista SP0 x64
VistaSP0x86                     - A Profile for Windows Vista SP0 x86
VistaSP1x64                     - A Profile for Windows Vista SP1 x64
VistaSP1x86                     - A Profile for Windows Vista SP1 x86
VistaSP2x64                     - A Profile for Windows Vista SP2 x64
VistaSP2x86                     - A Profile for Windows Vista SP2 x86
Win2003SP0x86                   - A Profile for Windows 2003 SP0 x86
Win2003SP1x64                   - A Profile for Windows 2003 SP1 x64
Win2003SP1x86                   - A Profile for Windows 2003 SP1 x86
Win2003SP2x64                   - A Profile for Windows 2003 SP2 x64
Win2003SP2x86                   - A Profile for Windows 2003 SP2 x86
Win2008R2SP0x64                 - A Profile for Windows 2008 R2 SP0 x64
Win2008R2SP1x64                 - A Profile for Windows 2008 R2 SP1 x64
Win2008SP1x64                   - A Profile for Windows 2008 SP1 x64
Win2008SP1x86                   - A Profile for Windows 2008 SP1 x86
Win2008SP2x64                   - A Profile for Windows 2008 SP2 x64
Win2008SP2x86                   - A Profile for Windows 2008 SP2 x86
Win7SP0x64                      - A Profile for Windows 7 SP0 x64
Win7SP0x86                      - A Profile for Windows 7 SP0 x86
Win7SP1x64                      - A Profile for Windows 7 SP1 x64
Win7SP1x86                      - A Profile for Windows 7 SP1 x86
WinXPSP1x64                     - A Profile for Windows XP SP1 x64
WinXPSP2x64                     - A Profile for Windows XP SP2 x64
WinXPSP2x86                     - A Profile for Windows XP SP2 x86
WinXPSP3x86                     - A Profile for Windows XP SP3 x86
```

```
vol.py --info | grep Linux
```

```
root@myubu:/tmp/dumpfile# vol.py --info | grep Linux
Volatile Systems Volatility Framework 2.3_beta
LinuxDebian-2_6_32-5-686x86 - A Profile for Linux Debian-2.6.32-5-686 x86
LinuxDebian5_26x86          - A Profile for Linux Debian5_26 x86
LinuxEvo4GARM               - A Profile for Linux Evo4G ARM
LinuxUbuntu1204_3_2_0_23x64 - A Profile for Linux Ubuntu1204_3_2_0_23 x64
LinuxUbuntu1204_3_2_0_32x64 - A Profile for Linux Ubuntu1204_3_2_0_32 x64
LinuxUbuntu1204_3_2_0_49x64 - A Profile for Linux Ubuntu1204_3_2_0_49 x64
LinuxUbuntu1204x64          - A Profile for Linux Ubuntu1204 x64
LinuxUbuntu1304_3_8_0_25x64 - A Profile for Linux Ubuntu1304_3_8_0_25 x64
LinuxUbuntu1304_3_8_0_27x64 - A Profile for Linux Ubuntu1304_3_8_0_27 x64
Linuxdfrws-profilex86       - A Profile for Linux dfrws-profile x86
linux_yarascan          - A shell in the Linux memory image
```

Festlegung der EXPORT Variablen:

```
export VOLATILITY_LOCATION=file:///tmp/stuxnet.vmem
export VOLATILITY_PROFILE=WinXPSP3x86
```

Ein Windows Profil ermitteln:

```
vol.py -f /tmp/zeus.vmem imageinfo
rekall -f /tmp/zeus.vmem imageinfo
```

```
root@myubu:/tmp/dumpfile# vol.py -f /tmp/zeus.vmem imageinfo
Volatile Systems Volatility Framework 2.3_beta
Determining profile based on KDBG search...

WARNING : volatility.obj      : Overlay structure tty_struct not present in vtypes
WARNING : volatility.obj      : Overlay structure tty_struct not present in vtypes
WARNING : volatility.obj      : Overlay structure tty_struct not present in vtypes
          Suggested Profile(s) : WinXPSP2x86, WinXPSP3x86
                     AS Layer1 : IA32PagedMemoryPae (Kernel AS)
                     AS Layer2 : FileAddressSpace (/tmp/zeus.vmem)
                      PAE type : PAE
                           DTB : 0x319000L
                          KDBG : 0x80544ce0
          Number of Processors : 1
     Image Type (Service Pack) : 2
                KPCR for CPU 0 : 0xffdff000
             KUSER_SHARED_DATA : 0xffdf0000
           Image date and time : 2010-08-15 19:17:56 UTC+0000
     Image local date and time : 2010-08-15 15:17:56 -0400
```

Netzwerk:

```
vol.py -f /tmp/zeus.vmem connscan
rekall -f /tmp/zeus.vmem connscan
```

```
root@myubu:/tmp/dumpfile# vol.py -f /tmp/zeus.vmem connscan
Volatile Systems Volatility Framework 2.3_beta
Offset(P)  Local Address             Remote Address            Pid
---------- ------------------------- ------------------------- ---
0x02214988 172.16.176.143:1054       193.104.41.75:80          856
0x06015ab0 0.0.0.0:1056              193.104.41.75:80          856
```

Registry:

`vol.py -f /tmp/zeus.vmem hivelist`

```
root@myubu:/tmp/dumpfile# vol.py -f /tmp/zeus.vmem hivelist
Volatile Systems Volatility Framework 2.3_beta
Virtual    Physical   Name
---------- ---------- ----
0xe1c49008 0x036dc008 \Device\HarddiskVolume1\Documents and Settings\LocalService\Local Settings\Application Dat
a\Microsoft\Windows\UsrClass.dat
0xe1c41b60 0x04010b60 \Device\HarddiskVolume1\Documents and Settings\LocalService\NTUSER.DAT
0xe1a39638 0x021eb638 \Device\HarddiskVolume1\Documents and Settings\NetworkService\Local Settings\Application D
ata\Microsoft\Windows\UsrClass.dat
0xe1a33008 0x01f98008 \Device\HarddiskVolume1\Documents and Settings\NetworkService\NTUSER.DAT
0xe153ab60 0x06b7db60 \Device\HarddiskVolume1\WINDOWS\system32\config\software
0xe1542008 0x06c48008 \Device\HarddiskVolume1\WINDOWS\system32\config\default
0xe1537b60 0x06ae4b60 \SystemRoot\System32\Config\SECURITY
0xe1544008 0x06c4b008 \Device\HarddiskVolume1\WINDOWS\system32\config\SAM
0xe13ae580 0x01bbd580 [no name]
0xe101b008 0x01867008 \Device\HarddiskVolume1\WINDOWS\system32\config\system
0xe1008978 0x01824978 [no name]
0xe1e158c0 0x009728c0 \Device\HarddiskVolume1\Documents and Settings\Administrator\Local Settings\Application Da
ta\Microsoft\Windows\UsrClass.dat
0xe1da4008 0x00f6e008 \Device\HarddiskVolume1\Documents and Settings\Administrator\NTUSER.DAT
root@myubu:/tmp/dumpfile#
```

Virtuelle Adresse für SAM: 0xe1544008
Virtuelle Adresse für system: 0xe101b008

Hashwerte für Rainbow Tables und Ophcrack:

`vol.py -f /tmp/zeus.vmem hashdump -y 0xe101b008 -s 0xe1544008`

```
root@myubu:/tmp/dumpfile# vol.py -f /tmp/zeus.vmem hashdump -y 0xe101b008 -s 0xe1544008
Volatile Systems Volatility Framework 2.3_beta
Administrator:500:e52cac67419a9a224a3b108f3fa6cb6d:8846f7eaee8fb117ad06bdd830b7586c:::
Guest:501:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
HelpAssistant:1000:4e857c004024e53cd538de64dedac36b:842b4013c45a3b8fec76ca54e5910581:::
SUPPORT_388945a0:1002:aad3b435b51404eeaad3b435b51404ee:8f57385a61425fc7874c3268aa249ea1:::
root@myubu:/tmp/dumpfile#
```

Software Hive für Run/WinLogon extrahieren:

RUN:

```
vol.py -f /tmp/zeus.vmem printkey -o 0xe153ab60 -K "Microsoft\
Windows\CurrentVersion\Run"
```

```
root@myubu:/tmp# vol.py -f /tmp/zeus.vmem printkey -o 0xe153ab60 -K "Microsoft\Windows\Curren
tVersion\Run"
Volatile Systems Volatility Framework 2.3_beta
Legend: (S) = Stable   (V) = Volatile

----------------------------
Registry: User Specified
Key name: Run (S)
Last updated: 2010-06-10 16:13:03 UTC+0000

Subkeys:

Values:
REG_SZ        VMware Tools    : (S) "C:\Program Files\VMware\VMware Tools\VMwareTray.exe"
REG_SZ        VMware User Process : (S) "C:\Program Files\VMware\VMware Tools\VMwareUser.exe"
```

RUNONCE:

```
vol.py -f /tmp/zeus.vmem printkey -o 0xe153ab60 -K "Microsoft\
Windows\CurrentVersion\RunOnce"
```

```
root@myubu:/tmp# vol.py -f /tmp/zeus.vmem printkey -o 0xe153ab60 -K "Microsoft\Windows\Curren
tVersion\RunOnce"
Volatile Systems Volatility Framework 2.3_beta
Legend: (S) = Stable   (V) = Volatile

----------------------------
Registry: User Specified
Key name: RunOnce (S)
Last updated: 2010-06-10 16:11:45 UTC+0000

Subkeys:

Values:
```

WINLOGON:

```
vol.py -f /tmp/zeus.vmem printkey -o 0xe153ab60 -K "Microsoft\
Windows NT\CurrentVersion\Winlogon"
```

```
root@myubu:/tmp# vol.py -f /tmp/zeus.vmem printkey -o 0xe153ab60 -K "Microsoft\Windows NT\CurrentVersion\Winlogon"
Volatile Systems Volatility Framework 2.3_beta
Legend: (S) = Stable   (V) = Volatile

----------------------------
Registry: User Specified
Key name: Winlogon (S)
Last updated: 2010-08-15 19:17:23 UTC+0000

Subkeys:
  (S) GPExtensions
  (S) Notify
  (S) SpecialAccounts
  (V) Credentials

Values:
REG_DWORD     AutoRestartShell : (S) 1
REG_SZ        DefaultDomainName : (S) BILLY-DB5B96DD3
REG_SZ        DefaultUserName : (S) Administrator
REG_SZ        LegalNoticeCaption : (S)
REG_SZ        LegalNoticeText : (S)
REG_SZ        PowerdownAfterShutdown : (S) 0
REG_SZ        ReportBootOk    : (S) 1
REG_SZ        Shell           : (S) Explorer.exe
REG_SZ        ShutdownWithoutLogon : (S) 0
REG_SZ        System          : (S)
REG_SZ        Userinit        : (S) C:\WINDOWS\system32\userinit.exe,C:\WINDOWS\system32\sdra64.exe,
REG_SZ        VmApplet        : (S) rundll32 shell32,Control_RunDLL "sysdm.cpl"
REG_DWORD     SfcQuota        : (S) 4294967295
REG_SZ        allocatecdroms  : (S) 0
REG_SZ        allocatedasd    : (S) 0
REG_SZ        allocatefloppies : (S) 0
REG_SZ        cachedlogonscount : (S) 10
REG_DWORD     forceunlocklogon : (S) 0
REG_DWORD     passwordexpirywarning : (S) 14
REG_SZ        scremoveoption  : (S) 0
REG_DWORD     AllowMultipleTSSessions : (S) 1
REG_EXPAND_SZ UIHost          : (S) logonui.exe
REG_DWORD     LogonType       : (S) 1
REG_SZ        Background      : (S) 0 0 0
REG_SZ        AutoAdminLogon  : (S) 0
REG_SZ        DebugServerCommand : (S) no
REG_DWORD     SFCDisable      : (S) 0
REG_SZ        WinStationsDisabled : (S) 0
REG_DWORD     HibernationPreviouslyEnabled : (S) 1
REG_DWORD     ShowLogonOptions : (S) 0
REG_SZ        AltDefaultUserName : (S) Administrator
REG_SZ        AltDefaultDomainName : (S) BILLY-DB5B96DD3
```

Prozesse:

```
vol.py -f /tmp/zeus.vmem pstree
rekall -f /tmp/zeus.vmem pstree
```

```
root@myubu:/tmp# vol.py -f /tmp/zeus.vmem pstree
Volatile Systems Volatility Framework 2.3_beta
Name                                                  Pid   PPid   Thds   Hnds Time
-------------------------------------------------- ------ ------ ------ ------ ----
 0x810b1660:System                                      4      0     58    379 1970-01-01 00:00:00 UTC+0000
. 0xff2ab020:smss.exe                                 544      4      3     21 2010-08-11 06:06:21 UTC+0000
.. 0xff1ec978:winlogon.exe                            632    544     24    536 2010-08-11 06:06:23 UTC+0000
... 0xff255020:lsass.exe                              688    632     21    405 2010-08-11 06:06:24 UTC+0000
... 0xff247020:services.exe                           676    632     16    288 2010-08-11 06:06:24 UTC+0000
.... 0xff1b8b28:vmtoolsd.exe                         1668    676      5    225 2010-08-11 06:06:35 UTC+0000
..... 0xff224020:cmd.exe                              124   1668      0 ------ 2010-08-15 19:17:55 UTC+0000
.... 0x80ff88d8:svchost.exe                           856    676     29    336 2010-08-11 06:06:24 UTC+0000
.... 0xff1d7da0:spoolsv.exe                          1432    676     14    145 2010-08-11 06:06:26 UTC+0000
.... 0x80fbf910:svchost.exe                          1028    676     88   1424 2010-08-11 06:06:24 UTC+0000
..... 0x80f60da0:wuauclt.exe                         1732   1028      7    189 2010-08-11 06:07:44 UTC+0000
..... 0x80f94588:wuauclt.exe                          468   1028      4    142 2010-08-11 06:09:37 UTC+0000
..... 0xff364310:wscntfy.exe                          888   1028      1     40 2010-08-11 06:06:49 UTC+0000
.... 0xff217560:svchost.exe                           936    676     11    288 2010-08-11 06:06:24 UTC+0000
.... 0xff143b28:TPAutoConnSvc.e                      1968    676      5    106 2010-08-11 06:06:39 UTC+0000
..... 0xff38b5f8:TPAutoConnect.e                     1084   1968      1     68 2010-08-11 06:06:52 UTC+0000
.... 0xff22d558:svchost.exe                          1088    676      7     93 2010-08-11 06:06:25 UTC+0000
.... 0xff218230:vmacthlp.exe                          844    676      1     37 2010-08-11 06:06:24 UTC+0000
.... 0xff25a7e0:alg.exe                               216    676      8    120 2010-08-11 06:06:39 UTC+0000
.... 0xff203b80:svchost.exe                          1148    676     15    217 2010-08-11 06:06:26 UTC+0000
.... 0xff1fdc88:VMUpgradeHelper                      1788    676      5    112 2010-08-11 06:06:38 UTC+0000
.. 0xff1ecda0:csrss.exe                               608    544     10    410 2010-08-11 06:06:23 UTC+0000
 0xff3865d0:explorer.exe                             1724   1708     13    326 2010-08-11 06:09:29 UTC+0000
. 0xff374980:VMwareUser.exe                           452   1724      8    207 2010-08-11 06:09:32 UTC+0000
. 0xff3667e8:VMwareTray.exe                           432   1724      1     60 2010-08-11 06:09:31 UTC+0000
```

Prozess ID 856 (aus z.B. Netzwerkverbindung → svchost.exe → Parent ID 676 → services

```
vol.py -f /tmp/zeus.vmem  procdump -D /tmp/dumpdir
rekall -f /tmp/zeus.vmem  procdump -D /tmp/dumpdir
```

```
root@myubu:/tmp# vol.py -f /tmp/zeus.vmem p 856 procexedump --dump-dir /tmp/dumpdir
Volatile Systems Volatility Framework 2.3_beta
Process(V) ImageBase  Name                 Result
---------- ---------- -------------------- ------
0x810b1660 ---------- System               Error: PEB at 0x0 is paged
0xff2ab020 0x48580000 smss.exe             Error: ImageBaseAddress at 0x48580000 is paged
0xff1ecda0 0x4a680000 csrss.exe            Error: ImageBaseAddress at 0x4a680000 is paged
0xff1ec978 0x01000000 winlogon.exe         OK: executable.632.exe
0xff247020 0x01000000 services.exe         OK: executable.676.exe
0xff255020 0x01000000 lsass.exe            OK: executable.688.exe
0xff218230 0x00400000 vmacthlp.exe         OK: executable.844.exe
0x80ff88d8 0x01000000 svchost.exe          OK: executable.856.exe
0xff217560 0x01000000 svchost.exe          OK: executable.936.exe
0x80fbf910 0x01000000 svchost.exe          OK: executable.1028.exe
0xff22d558 0x01000000 svchost.exe          OK: executable.1088.exe
0xff203b80 0x01000000 svchost.exe          OK: executable.1148.exe
0xff1d7da0 0x01000000 spoolsv.exe          OK: executable.1432.exe
0xff1b8b28 0x00400000 vmtoolsd.exe         OK: executable.1668.exe
0xff1fdc88 0x00400000 VMUpgradeHelper      OK: executable.1788.exe
0xff143b28 0x00400000 TPAutoConnSvc.e      OK: executable.1968.exe
0xff25a7e0 0x01000000 alg.exe              OK: executable.216.exe
0xff364310 0x01000000 wscntfy.exe          OK: executable.888.exe
0xff38b5f8 0x00400000 TPAutoConnect.e      OK: executable.1084.exe
0x80f60da0 0x00400000 wuauclt.exe          OK: executable.1732.exe
0xff3865d0 0x01000000 explorer.exe         OK: executable.1724.exe
0xff3667e8 0x00400000 VMwareTray.exe       OK: executable.432.exe
0xff374980 0x00400000 VMwareUser.exe       OK: executable.452.exe
0x80f94588 0x00400000 wuauclt.exe          OK: executable.468.exe
0xff224020 ---------- cmd.exe              Error: PEB at 0x7ffdf000 is paged
```

```
file dumpdir/executable.856.exe
```

```
root@myubu:/tmp# file dumpdir/executable.856.exe
dumpdir/executable.856.exe: PE32 executable (GUI) Intel 80386, for MS Windows
root@myubu:/tmp#
```

auf www.virustotal.com:

Sophos	✓	20130624
SUPERAntiSpyware	✓	20130624
Symantec	WS.Reputation.1	20130624
TheHacker	✓	20130624
TotalDefense	✓	20130624
TrendMicro	✓	20130624
TrendMicro-HouseCall	TROJ_GEN.F47V1024	20130624
VBA32	✓	20130624
VIPRE	✓	20130624

Malware finden:

```
vol.py -f /tmp/zeus.vmem malfind -D /tmp/zeusmalware/
rekall -f /tmp/zeus.vmem malfind -D /tmp/zeusmalware/
```

```
root@myubu:/tmp# ll zeusmalware/
insgesamt 4856
drwxr-xr-x  2 root root    4096 Sep 26 14:05 ./
drwxrwxrwt 14 root root    4096 Sep 26 14:05 ../
-rw-r--r--  1 root root  155648 Sep 26 14:05 process.0x80f60da0.0x1000000.dmp
-rw-r--r--  1 root root    4096 Sep 26 14:05 process.0x80f60da0.0x2800000.dmp
-rw-r--r--  1 root root  155648 Sep 26 14:05 process.0x80f94588.0x12d0000.dmp
-rw-r--r--  1 root root    4096 Sep 26 14:05 process.0x80f94588.0x1410000.dmp
-rw-r--r--  1 root root    4096 Sep 26 14:05 process.0x80fbf910.0x1f70000.dmp
-rw-r--r--  1 root root  155648 Sep 26 14:05 process.0x80fbf910.0x2450000.dmp
-rw-r--r--  1 root root  155648 Sep 26 14:05 process.0x80ff88d8.0xb70000.dmp
-rw-r--r--  1 root root    4096 Sep 26 14:05 process.0x80ff88d8.0xcb0000.dmp
-rw-r--r--  1 root root  155648 Sep 26 14:05 process.0x810b1660.0x170000.dmp
-rw-r--r--  1 root root  155648 Sep 26 14:05 process.0x810b1660.0x1a0000.dmp
-rw-r--r--  1 root root  155648 Sep 26 14:05 process.0x810b1660.0x1d0000.dmp
-rw-r--r--  1 root root  155648 Sep 26 14:05 process.0xff143b28.0xdf0000.dmp
-rw-r--r--  1 root root    4096 Sep 26 14:05 process.0xff143b28.0xeb0000.dmp
-rw-r--r--  1 root root  155648 Sep 26 14:05 process.0xff1b8b28.0x15e0000.dmp
-rw-r--r--  1 root root    4096 Sep 26 14:05 process.0xff1b8b28.0x16a0000.dmp
-rw-r--r--  1 root root  155648 Sep 26 14:05 process.0xff1d7da0.0x920000.dmp
-rw-r--r--  1 root root    4096 Sep 26 14:05 process.0xff1d7da0.0xb60000.dmp
-rw-r--r--  1 root root   16384 Sep 26 14:05 process.0xff1ec978.0x2c930000.dmp
-rw-r--r--  1 root root   16384 Sep 26 14:05 process.0xff1ec978.0x33470000.dmp
-rw-r--r--  1 root root   16384 Sep 26 14:05 process.0xff1ec978.0x37ec0000.dmp
-rw-r--r--  1 root root   16384 Sep 26 14:05 process.0xff1ec978.0x71ee0000.dmp
-rw-r--r--  1 root root   16384 Sep 26 14:05 process.0xff1ec978.0x78850000.dmp
-rw-r--r--  1 root root   16384 Sep 26 14:05 process.0xff1ec978.0x793e0000.dmp
-rw-r--r--  1 root root  155648 Sep 26 14:05 process.0xff1ec978.0xae0000.dmp
-rw-r--r--  1 root root 1048576 Sep 26 14:05 process.0xff1ecda0.0x7f6f0000.dmp
-rw-r--r--  1 root root  155648 Sep 26 14:05 process.0xff1fdc88.0x930000.dmp
```

YaraScan:

Stuxnet nutzt Signaturen von Realtek und jmicron !

Eigene Yara Regeln anlegen:

stolen_signatures.yar :

```
rule stolen_signatures
{
strings:
$find1 = "realtek" nocase
$find2 = "jmicron" nocase
condition:
$find1 or $find2
```

}

```
vol.py -f stuxnet.vmem yarascan
--yara-file=/tmp/stolen_signatures.yar  | grep -A10 1928
```

```
root@myubu:/tmp# vol.py -f stuxnet.vmem yarascan --yara-file=/tmp/dumpfile/stolen_signatures.yar  | grep -a10 1928
Volatile Systems Volatility Framework 2.3_beta
0x00e184a8  75 63 74 6f 72 20 43 6f 72 70 30 81 9f 30 0d 06   uctor.Corp0..0..
0x00e184b8  09 2a 86 48 86 f7 0d 01 01 01 05 00 03 81 8d 00   .*.H...........
0x00e184c8  30 81 89 02 81 81 00 b5 6c 5e 03 88 79 57 9a 4f   0.......l^..yW.O
Rule: stolen_signatures
Owner: Process svchost.exe Pid 940
0x00e18987  72 65 61 6c 74 65 6b 2e 63 6f 6d 30 0d 06 09 2a   realtek.com0...*
0x00e18997  86 48 86 f7 0d 01 01 01 05 00 04 81 80 a4 49 5b   .H...........I[
0x00e189a7  37 42 66 ec 7a 06 d8 38 97 f5 74 d4 ed 69 70 33   7Bf.z..8..t..ip3
0x00e189b7  95 ea f0 5a 04 f3 75 3a 1d 3d a7 e2 28 1d 74 ba   ...Z..u:.=..(.t.
Rule: stolen_signatures
Owner: Process lsass.exe Pid 1928
0x00988424  52 65 61 6c 74 65 6b 20 53 65 6d 69 63 6f 6e 64   Realtek.Semicond
0x00988434  75 63 74 6f 72 20 43 6f 72 70 31 3e 30 3c 06 03   uctor.Corp1>0<..
0x00988444  55 04 0b 13 35 44 69 67 69 74 61 6c 20 49 44 20   U...5Digital.ID.
0x00988454  43 6c 61 73 73 20 33 20 2d 20 4d 69 63 72 6f 73   Class.3.-.Micros
Rule: stolen_signatures
Owner: Process lsass.exe Pid 1928
0x00988498  52 65 61 6c 74 65 6b 20 53 65 6d 69 63 6f 6e 64   Realtek.Semicond
0x009884a8  75 63 74 6f 72 20 43 6f 72 70 30 81 9f 30 0d 06   uctor.Corp0..0..
0x009884b8  09 2a 86 48 86 f7 0d 01 01 01 05 00 03 81 8d 00   .*.H...........
0x009884c8  30 81 89 02 81 81 00 b5 6c 5e 03 88 79 57 9a 4f   0.......l^..yW.O
Rule: stolen_signatures
Owner: Process lsass.exe Pid 1928
0x00988987  72 65 61 6c 74 65 6b 2e 63 6f 6d 30 0d 06 09 2a   realtek.com0...*
0x00988997  86 48 86 f7 0d 01 01 01 05 00 04 81 80 a4 49 5b   .H...........I[
0x009889a7  37 42 66 ec 7a 06 d8 38 97 f5 74 d4 ed 69 70 33   7Bf.z..8..t..ip3
0x009889b7  95 ea f0 5a 04 f3 75 3a 1d 3d a7 e2 28 1d 74 ba   ...Z..u:.=..(.t.
```

Malware nutzt häufig das UPX Komprimierungsformat:

upx_find.yar :

rule upx_find
{
strings:
$upx0 = "upx0" nocase
$upx1 = "upx1" nocase
$rsrc = ".rsrc" nocase
condition:
$upx0 and $upx1 and $rsrc
}

```
vol.py -f stuxnet.vmem yarascan --yara-file=/tmp/upx_find.yar  |
grep -A10 1928
```

```
root@myubu:/tmp# vol.py -f stuxnet.vmem yarascan --yara-file=/tmp/dumpfile/upx_find.yar  | grep -a10 1928
Volatile Systems Volatility Framework 2.3_beta
0x00080238  00 8e 07 00 00 04 00 00 00 00 00 00 00 00 00 00   ................
0x00080248  00 00 00 00 40 00 00 e0 2e 72 73 72 63 00 00 00   ....@....rsrc...
0x00080258  00 10 00 00 00 70 13 00 00 08 00 00 00 92 07 00   .....p..........
Rule: upx_find
Owner: Process lsass.exe Pid 868
0x00080250  2e 72 73 72 63 00 00 00 00 10 00 00 00 70 13 00   .rsrc........p..
0x00080260  00 08 00 00 00 92 07 00 00 00 00 00 00 00 00 00   ................
0x00080270  00 00 00 00 40 00 00 c0 00 00 00 00 00 00 00 00   ....@...........
0x00080280  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00   ................
Rule: upx_find
Owner: Process lsass.exe Pid 1928
0x00080200  55 50 58 30 00 00 00 00 00 d0 0b 00 00 10 00 00   UPX0............
0x00080210  00 00 00 00 00 04 00 00 00 00 00 00 00 00 00 00   ................
0x00080220  00 00 00 00 80 00 00 e0 55 50 58 31 00 00 00 00   ........UPX1....
0x00080230  00 90 07 00 00 e0 0b 00 00 8e 07 00 00 04 00 00   ................
Rule: upx_find
Owner: Process lsass.exe Pid 1928
0x00080228  55 50 58 31 00 00 00 00 00 90 07 00 00 e0 0b 00   UPX1............
0x00080238  00 8e 07 00 00 04 00 00 00 00 00 00 00 00 00 00   ................
0x00080248  00 00 00 00 40 00 00 e0 2e 72 73 72 63 00 00 00   ....@....rsrc...
0x00080258  00 10 00 00 00 70 13 00 00 08 00 00 00 92 07 00   .....p..........
Rule: upx_find
Owner: Process lsass.exe Pid 1928
0x00080250  2e 72 73 72 63 00 00 00 00 10 00 00 00 70 13 00   .rsrc........p..
0x00080260  00 08 00 00 00 92 07 00 00 00 00 00 00 00 00 00   ................
0x00080270  00 00 00 00 40 00 00 c0 00 00 00 00 00 00 00 00   ....@...........
0x00080280  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00   ................
Rule: upx_find
```

Stuxnet startet einen eigenen RPC Dienst zur Weiterverbreitung:

```
vol.py -f stuxnet.vmem dlllist -p 868 1928
```

```
root@myubu:/tmp# vol.py -f stuxnet.vmem dlllist -p 868 1928
Volatile Systems Volatility Framework 2.3_beta
************************************************************************
lsass.exe pid:    868
Command line : "C:\WINDOWS\\system32\\lsass.exe"
Service Pack 3

Base             Size  LoadCount Path
---------- ---------- ---------- ----
0x01000000     0x6000     0xffff C:\WINDOWS\system32\lsass.exe
0x7c900000    0xaf000     0xffff C:\WINDOWS\system32\ntdll.dll
0x7c800000    0xf6000     0xffff C:\WINDOWS\system32\kernel32.dll
0x77dd0000    0x9b000     0xffff C:\WINDOWS\system32\ADVAPI32.dll
0x77e70000    0x92000     0xffff C:\WINDOWS\system32\RPCRT4.dll
0x77fe0000    0x11000     0xffff C:\WINDOWS\system32\Secur32.dll
0x7e410000    0x91000     0xffff C:\WINDOWS\system32\USER32.dll
0x77f10000    0x49000     0xffff C:\WINDOWS\system32\GDI32.dll
```

und ebenfalls für die PID 1928

Timeline:

Mit den drei Plugins timeline, mftparser und shellbag kann eine Timeline erzeugt werden, die wichtige Zeitstempel über Prozesse und Registry Einträge erstellt.

```
vol.py --plugins=contrib/plugins -f stuxnet.vmem timeliner --output=body --
output-file=timeliner.txt -R
vol.py -f stuxnet.vmem mftparser -C --output=body --output-file=mft.txt
vol.py -f stuxnet.vmem shellbags --output=body –output-file=shellbags.txt
```

Die Dateien liegen im Sleuthkit bodyfile Format vor und werden noch zusammengefasst:

```
cat timeliner.txt mft.txt shellbags.txt >> bodyfile.txt
```

Die Konvertierung erfolgt mit dem mactime Programm:

```
mactime -b bodyfile.txt > mactime.txt
```

http://volatility-labs.blogspot.de/2013/05/movp-ii-23-creating-timelines-with.html

Internet Explorer (von Patrick Zimmermann DHBW)

```
vol.py -f iexplore.raw --profile=WinXPSP3x86 yarascan -Y "/(URL
|REDR|LEAK)/" -p 1272
```

oder nach „www.“:

```
vol.py -f iexplore.raw --profile=WinXPSP3x86 yarascan -Y "www."
-p 1272
```

oder:

```
vol.py -f iexplore.raw --profile=WinXPSP3x86 yarascan -Y
"ftp://" -p 1272
```

oder:

```
vol.py -f iexplore.raw  --profile=WinXPSP3x86 yarascan -Y
"http://" -p 1272
```

```
root@hpm:/tmp# vol.py -f memdump.raw yarascan -Y "http://" -p 1272
Volatility Foundation Volatility Framework 2.3.1
Rule: r1
Owner: Process IEXPLORE.EXE Pid 1272
0x00120e32  68 74 74 70 3a 2f 2f 64 65 2e 77 69 6b 69 70 65   http://de.wikipe
0x00120e42  64 69 61 2e 6f 72 67 2f 77 69 6b 69 2f 46 6f 72   dia.org/wiki/For
0x00120e52  65 6e 73 69 6b 00 00 00 00 00 00 00 00 00 00 00   ensik...........
0x00120e62  00 00 00 00 00 00 00 00 00 00 00 00 00 00 5c 03   ..............\.
Rule: r1
Owner: Process IEXPLORE.EXE Pid 1272
0x00122752  68 74 74 70 3a 2f 2f 64 65 2e 77 69 6b 69 70 65   http://de.wikipe
0x00122762  64 69 61 2e 6f 72 67 2f 77 69 6b 69 2f 46 6f 72   dia.org/wiki/For
0x00122772  65 6e 73 69 6b 00 10 00 02 00 00 00 00 10 00 00   ensik...........
0x00122782  00 00 00 00 00 00 34 00 10 1f 46 00 6f 00 72 00   ......4...F.o.r.
Rule: r1
Owner: Process IEXPLORE.EXE Pid 1272
0x00181fc3  68 74 74 70 3a 2f 2f 77 77 77 2e 63 65 72 74 70   http://www.certp
0x00181fd3  6c 75 73 2e 63 6f 6d 2f 43 52 4c 2f 63 6c 61 73   lus.com/CRL/clas
0x00181fe3  73 33 54 53 2e 63 72 6c 30 0d 06 09 2a 86 48 86   s3TS.crl0...*.H.
0x00181ff3  f7 0d 01 01 05 05 00 03 82 01 01 00 b5 9f 68 9c   ..............h.
Rule: r1
```

Virtualbox Memory Dump

Virtualbox besitzt leider keine so elegante Möglichkeit wie VMware um einen Volatility gerechten Memory Dump zu erzeugen. Mit wenigen Kommandozeilen Befehlen ist dies aber schnell erreicht:

```
VBoxManage list vms
```

listet die Namen der Gastmaschinen. Wichtig dabei! Den Befehl als User absetzen, da die VM mit User Rechten gestartet wird.

```
VBoxManage debugvm imagefile dumpcore --filename imagefile.dump
```

Um den Dump mit Volatility nutzen zu können muss das Format noch konvertiert werden:

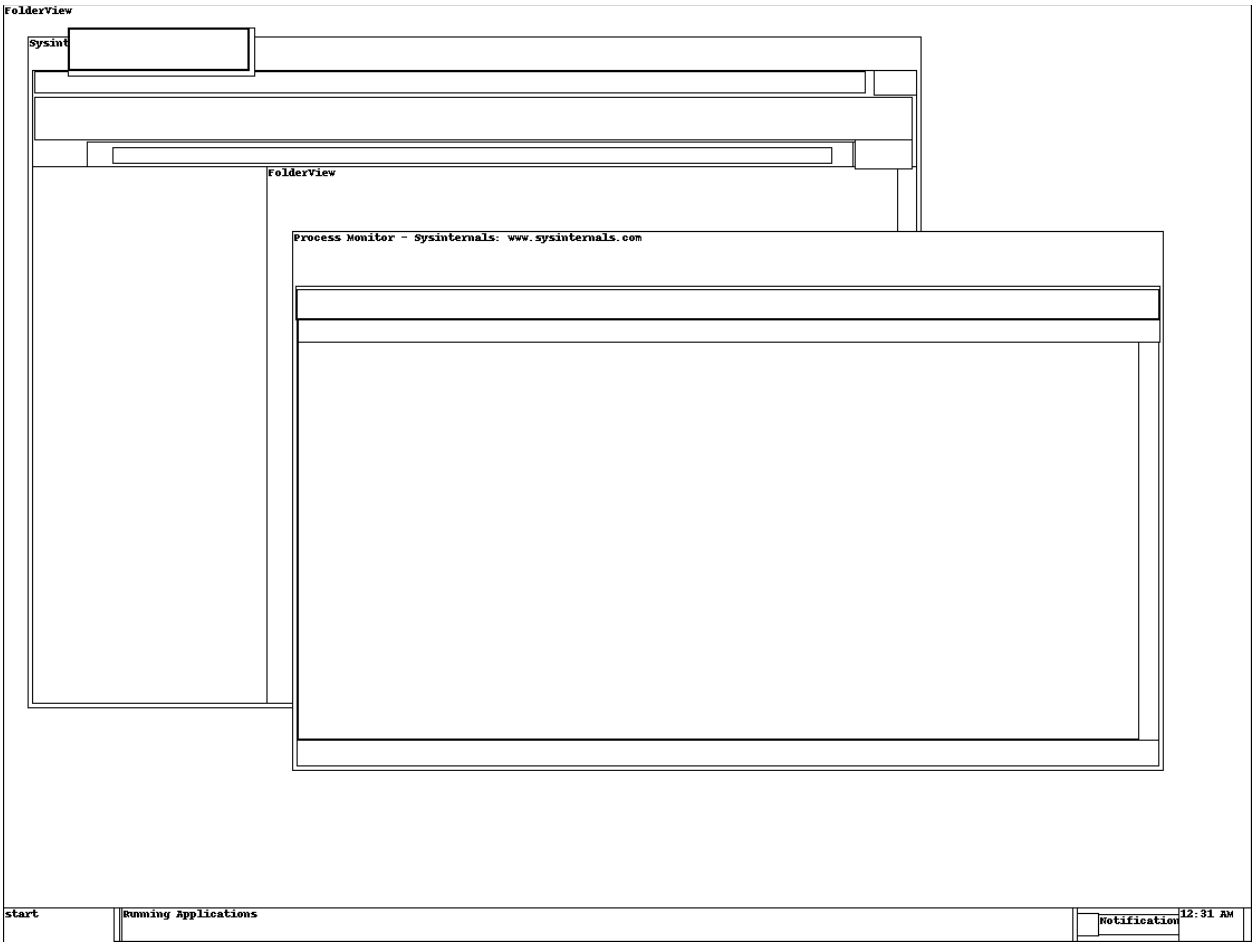
```
vol.py -f imagefile.dump  imagecopy -O copy.raw
```

Der Befehl imagecopy funktioniert nur bei Windows Gästen. Linux und Mac Gäste können derzeit nicht konvertiert werden!

Das Screenshot Plugin

```
vol.py -f stuxnet.vmem screenshot -D shots/
Volatility Foundation Volatility Framework 2.3.1
Wrote shots/session_0.Service-0x0-3e7$.Default.png
Wrote shots/session_0.Service-0x0-3e4$.Default.png
Wrote shots/session_0.Service-0x0-3e5$.Default.png
Wrote shots/session_0.SAWinSta.SADesktop.png
```

```
Wrote shots/session_0.WinSta0.Default.png
Wrote shots/session_0.WinSta0.Disconnect.png
Wrote shots/session_0.WinSta0.Winlogon.png
```




```
vol.py -f stuxnet.vmem dumpcerts -D certs --ssl
Volatility Foundation Volatility Framework 2.3.1
Pid      Process          Address    Type                Length   File
Subject
-------- ---------------- ---------- -------------------- -------- ------------------------
-------
     624 winlogon.exe     0x010165d8 _X509_PUBLIC_CERT        1042 624-10165d8.crt
Microsoft Corporation
     668 services.exe     0x000b6b20 _X509_PUBLIC_CERT         576 668-b6b20.crt
VeriSign/VeriSign Commercial Software Publishers CA
     668 services.exe     0x000b6d70 _X509_PUBLIC_CERT         433 668-b6d70.crt
     668 services.exe     0x000c8268 _X509_PUBLIC_CERT         665 668-c8268.crt
Fundacion FESTE
     668 services.exe     0x000decb0 _X509_PUBLIC_CERT         642 668-decb0.crt
Equifax Secure Inc.
     668 services.exe     0x000e3f90 _X509_PUBLIC_CERT         787 668-e3f90.crt
Thawte Consulting cc/Certification Services Division
     668 services.exe     0x000e49d8 _X509_PUBLIC_CERT         785 668-e49d8.crt
SecureNet CA Root
     668 services.exe     0x000e4cf8 _X509_PUBLIC_CERT         761 668-e4cf8.crt            FNMT/
FNMT Clase 2 CA
     668 services.exe     0x000e5000 _X509_PUBLIC_CERT         780 668-e5000.crt
Unizeto Sp. z o.o.
```

```
     668 services.exe     0x000e77b8 _X509_PUBLIC_CERT         506 668-e77b8.crt            GTE
Corporation
     668 services.exe     0x000ec370 _X509_PUBLIC_CERT        1181 668-ec370.crt
Microsoft Corporation/nCipher DSE ESN:D8A9-CFCC-579C
     668 services.exe     0x000ecfc0 _X509_PUBLIC_CERT        1200 668-ecfc0.crt
Microsoft Corporation
     668 services.exe     0x000f0d48 _X509_PUBLIC_CERT         917 668-f0d48.crt
Certplus
     668 services.exe     0x000f1968 _X509_PUBLIC_CERT         700 668-f1968.crt
VeriSign/VeriSign Time Stamping Service Root/NO LIABILITY ACCEPTED
     668 services.exe     0x000f1fc8 _X509_PUBLIC_CERT         915 668-f1fc8.crt
Certiposte
     668 services.exe     0x000f2408 _X509_PUBLIC_CERT         685 668-f2408.crt
Microsoft Corporation/Microsoft Time Stamping Service Root/Copyright (c) 1997 Microsoft Corp.
     668 services.exe     0x000f29e0 _X509_PUBLIC_CERT         927 668-f29e0.crt
Deutsche Telekom AG/T-TeleSec Trust Center
     668 services.exe     0x000f35f8 _X509_PUBLIC_CERT         982 668-f35f8.crt            MSFT
     668 services.exe     0x000f3bb8 _X509_PUBLIC_CERT         870 668-f3bb8.crt
VeriSign Trust Network/www.verisign.com/repository/RPA Incorp. By Ref.
     668 services.exe     0x000f3fe8 _X509_PUBLIC_CERT        1042 668-f3fe8.crt
Microsoft Corporation
     668 services.exe     0x000f4408 _X509_PUBLIC_CERT         879 668-f4408.crt
Thawte Consulting cc/Certification Services Division
     668 services.exe     0x000f4808 _X509_PUBLIC_CERT         612 668-f4808.crt
Certisign Certificadora Digital Ltda./Certisign Autoridade Certificadora AC1S
     668 services.exe     0x000f4a90 _X509_PUBLIC_CERT        1433 668-f4a90.crt
     668 services.exe     0x000f5798 _X509_PUBLIC_CERT         809 668-f5798.crt            Japan
Certification Services
     668 services.exe     0x000f5b58 _X509_PUBLIC_CERT        1140 668-f5b58.crt            The
USERTRUST Network/http://www.usertrust.com
     668 services.exe     0x000f60b8 _X509_PUBLIC_CERT        1078 668-f60b8.crt
AddTrust AB/AddTrust External TTP Network
```

vol.py -f stuxnet.vmem dumpcerts -D certs –ssl | grep -i realtek

```
Volatility Foundation Volatility Framework 2.3.1
     668 services.exe     0x015082e1 _X509_PUBLIC_CERT        1286 668-15082e1.crt
Realtek Semiconductor Corp/Digital ID Class 3 - Microsoft Software Validation v2/RTCN
     940 svchost.exe      0x00e182e1 _X509_PUBLIC_CERT        1286 940-e182e1.crt
Realtek Semiconductor Corp/Digital ID Class 3 - Microsoft Software Validation v2/RTCN
    1928 lsass.exe        0x009882e1 _X509_PUBLIC_CERT        1286 1928-9882e1.crt
Realtek Semiconductor Corp/Digital ID Class 3 - Microsoft Software Validation v2/RTCN
```

openssl x509 -inform der -in certs/668-15082e1.crt -text

```
Certificate:
    Data:
        Version: 3 (0x2)
        Serial Number:
            5e:6d:dc:87:37:50:82:84:58:14:f4:42:d1:d8:2a:25
    Signature Algorithm: sha1WithRSAEncryption
        Issuer: C=US, O=VeriSign, Inc., OU=VeriSign Trust Network, OU=Terms of use at https://www.verisign.com/rpa (c)04,
CN=VeriSign Class 3 Code Signing 2004 CA
        Validity
            Not Before: Mar 15 00:00:00 2007 GMT
            Not After : Jun 11 23:59:59 2010 GMT
        Subject: C=TW, ST=Taiwan, L=Hsinchu, O=Realtek Semiconductor Corp, OU=Digital ID Class 3 - Microsoft Software Validation
v2, OU=RTCN, CN=Realtek Semiconductor Corp
        Subject Public Key Info:
            Public Key Algorithm: rsaEncryption
                Public-Key: (1024 bit)
                Modulus:
                    00:b5:6c:5e:03:88:79:57:9a:4f:ce:02:30:b9:05:
                    1f:06:13:7a:57:17:4e:ee:59:d6:b0:20:d0:7c:e9:
                    17:32:bb:03:e8:13:45:1a:3e:22:57:4c:b4:bc:b3:
                    c5:3d:32:62:e8:a9:6c:f2:03:d1:72:ee:1e:f2:65:
                    ce:dd:df:0f:6e:77:08:a7:b2:13:92:db:30:f3:d8:
                    8e:b9:53:46:d5:08:16:4f:6d:92:a5:f2:74:5e:d9:
                    42:d2:89:50:92:53:0a:e6:ff:67:3f:14:59:fe:0d:
                    af:e0:97:8e:1f:f0:eb:6f:c6:e1:93:3e:5e:50:a6:
                    26:65:69:21:84:32:be:a3:27
                Exponent: 65537 (0x10001)
        X509v3 extensions:
            X509v3 Basic Constraints:
                CA:FALSE
            X509v3 Key Usage: critical
                Digital Signature
            X509v3 CRL Distribution Points:

                Full Name:
                  URI:http://CSC3-2004-crl.verisign.com/CSC3-2004.crl
```

```
            X509v3 Certificate Policies:
                Policy: 2.16.840.1.113733.1.7.23.3
                  CPS: https://www.verisign.com/rpa

            X509v3 Extended Key Usage:
                Code Signing
            Authority Information Access:
                OCSP - URI:http://ocsp.verisign.com
                CA Issuers - URI:http://CSC3-2004-aia.verisign.com/CSC3-2004-aia.cer

            X509v3 Authority Key Identifier:
                keyid:08:F5:51:E8:FB:FE:3D:3D:64:36:7C:68:CF:5B:78:A8:DF:B9:C5:37

            Netscape Cert Type:
                Object Signing
            1.3.6.1.4.1.311.2.1.27:
                0.......
    Signature Algorithm: sha1WithRSAEncryption
         bd:ba:23:02:36:e9:76:e9:cb:22:7a:ef:53:7b:3f:66:43:92:
         01:43:e7:90:94:b4:52:d7:2c:89:82:24:39:fe:3f:7b:24:65:
         d0:c2:72:56:18:a4:f6:0b:d3:9b:75:f1:63:62:89:5b:5d:2e:
         9e:f8:1a:51:48:6b:b8:e6:cd:96:b7:f1:32:31:b2:a1:cd:e3:
         bd:d9:04:7b:28:a9:5c:bf:be:3e:f3:e6:03:52:2c:75:88:5a:
         1a:ca:99:de:b9:3f:99:77:10:db:63:ed:68:b4:09:fb:ee:8d:
         c3:6a:c3:82:ec:12:e8:1a:22:61:28:8a:f6:4e:5d:1a:c3:50:
         97:3b:22:da:99:ab:3f:38:9b:4d:d1:da:01:e3:40:14:aa:a5:
         9d:13:1c:71:ea:cb:99:99:52:65:f5:94:af:d7:d0:ca:c7:04:
         20:eb:b7:41:4b:82:cc:07:84:de:6f:2a:51:8f:7b:0a:25:db:
         f0:1d:09:56:cc:30:5b:a2:55:5b:6d:da:8e:d6:a7:dd:1d:c6:
         ff:3d:74:f8:ff:e2:60:35:66:1f:a8:66:72:c1:fb:76:46:ca:
         df:24:ac:1a:66:fa:11:41:b7:f4:51:89:dc:da:49:99:3c:3d:
         d5:77:2d:b3:d3:d3:64:50:18:f4:95:f4:dd:9f:8d:76:41:60:
         25:b4:70:8b
-----BEGIN CERTIFICATE-----
MIIFBjCCA+6gAwIBAgIQXm3chzdQgoRYFPRC0dgqJTANBgkqhkiG9w0BAQUFADCB
tDELMAkGA1UEBhMCVVMxFzAVBgNVBAoTDlZlcmlTaWduLCBJbmMuMR8wHQYDVQQL
ExZWZXJpU2lnbiBUcnVzdCBOZXR3b3JrMTswOQYDVQQLEzJUZXJtcyBvZiB1c2Ug
YXQgaHR0cHM6Ly93d3cudmVyaXNpZ24uY29tL3JwYSAoYykwNDEuMCwGA1UEAxMl
VmVyaVNpZ24gQ2xhc3MgMyBDb2RlIFNpZ25pbmcgMjAwNCBDQTAeFw0wNzAzMTUw
MDAwMDBaFw0xMDA2MTEyMzU5NTlaMIHJMQswCQYDVQQGEwJUVzEPMA0GA1UECBMG
VGFpd2FuMRAwDgYDVQQHEwdIc2luY2h1MSMwIQYDVQQKFBpSZWFsdGVrIFNlbWlj
b25kdWN0b3IgQ29ycDE+MDwGA1UECxM1RGlnaXRhbCBJRCBDbGFzcyAzIC0gTWlj
cm9zb2Z0IFNvZnR3YXJlIFZhbGlkYXRpb24gdjIxDTALBgNVBAsUBFJUQ04xIzAh
BgNVBAMUGlJlYWx0ZWsgU2VtaWNvbmRlY3RvciBDb3JwMIGfMA0GCSqGSIb3DQEB
AQUAA4GNADCBiQKBgQC1bF4DiH1Xmk/OAjC5BR8GE3pXF07uWdawINB86RcyuwPo
E0UaPiJXTLS8s8U9MmLoqWzyA9Fy7h7yZc7d3w9udwinshOS2zDz2I65U0bVCBZP
bZKl8nRe2ULSiVCSUwrm/2c/FFn+Da/gl44f8OtvxuGTPl5QpiZlaSGEMr6jJwID
AQABo4IBfzCCAXswCQYDVR0TBAIwADAOBgNVHQ8BAf8EBAMCB4AwQAYDVR0fBDkw
NzA1oDOgMYYvaHR0cDovL0NTQzMtMjAwNC1jcmwudmVyaXNpZ24uY29tL0NTQzMt
MjAwNC5jcmwwRAYDVR0gBD0wOzA5Bgtghkg BhvhFAQcXAzAqMCgGCCsGAQUFBwIB
FhxodHRwczovL3d3dy52ZXJpc2lnbi5jb20vcnBhMBMGA1UdJQQMMAoGCCsGAQUF
BwMDMHUGCCsGAQUFBwEBBGkwZzAkBggrBgEFBQcwAYYYaHR0cDovL29jc3AudmVy
aXNpZ24uY29tMD8GCCsGAQUFBzAChjNodHRwOi8vQ1NDMy0yMDA0LWFpYS52ZXJp
c2lnbi5jb20vQ1NDMy0yMDA0LWFpYS5jZXIwHwYDVR0jBBgwFoAUCPVR6Pv+PT1k
Nnxoz1t4qN+5xTcwEQYJYIZIAYb4QgEBBAQDAgQQMBYGCisGAQQBgjcCARsECDAG
AQEAAQH/MA0GCSqGSIb3DQEBBQUAA4IBAQC9uiMCNul26csieu9Tez9mQ5IBQ+eQ
lLRS1yyJgiQ5/j97JGXQwnJWGKT2C9ObdfFjYolbXS6e+BpRSGu45s2Wt/EyMbKh
ze092QR7KKlcv74+8+YDUix1iFoaypneuT+ZdxDbY+1otAn77o3DasOC7BLoGiJh
KIr2Tl0aw1CXOyLamas/OJtN0doB40AUqqWdExxx6suZmVJl9ZSv19DKxwQg67dB
S4LMB4TebypRj3sKJdvwHQlWzDBbolVbbdqO1qfdHcb/PXT4/+JgNWYfqGZywft2
RsrfJKwaZvoRQbf0UYnc2kmZPD3Vdy2z09NkUBj01fTdn412QWAltHCL
-----END CERTIFICATE-----
```

Truecrypt:

Das alte Plugin cryptoscan ist nicht mehr kompatibel mit den aktuellen Volatility Versionen. Mit Version 2.4 wird ein neues Plugin **truecryptsummary** mitgeliefert, welches AES Schlüssel findet. Derzeit sind lediglich begrenzte Möglichkeiten vorhanden um Truecrypt zu analysieren/cracken.

Das folgende Beispiel prüft auf ein gestartetes Truecrypt und ermittelt den Laufwerksbezeichner:

```
vol.py -f truecrypt.raw pslist | grep -i truecrypt
Volatility Foundation Volatility Framework 2.3.1
0x88d2d020 TrueCrypt.exe          2272   1716      3      144      0      0 2014-05-10 10:37:42
UTC+0000
```

```
vol.py -f truecrypt.raw symlinkscan | grep -i truecrypt
Volatility Foundation Volatility Framework 2.3.1
0x2b5619d8      1      0 2014-05-10 10:37:28 UTC+0000   TrueCrypt            \Device\TrueCrypt
0x333a1770      1      0 2014-05-10 10:39:32 UTC+0000   F:                   \Device\
TrueCryptVolumeF
```

```
0x39b3ec70      1      0 2014-05-10 10:39:32 UTC+0000   Volume{14...7bc3824} \Device\
TrueCryptVolumeF
0x3cf0b818      1      0 2014-05-10 10:39:32 UTC+0000   F:                   \Device\
TrueCryptVolumeF
```

Die AES Keys von Truecrypt können mit aeskeyfind ermittelt werden:

```
./aeskeyfind  /tmp/autopsy.dump
a6d6dbdf4922f83ce680f8c0813cbb6654f037545f939c29288b0b0f222fe2a5
388ce276c71ffe30cecc7a2f87f7271b365024acf37e39bb5d2f7b2b9487a01e
000102030405060708090a0b0c0d0e0f101112131415161718191a1b1c1d1e1f
000102030405060708090a0b0c0d0e0f101112131415161718191a1b1c1d1e1f
Keyfind progress: 100%
```

Erweiterte Information liefert die -qv Option:

```
./aeskeyfind -qv /tmp/autopsy.dump
FOUND POSSIBLE 256-BIT KEY AT BYTE 8f066f8

KEY: a6d6dbdf4922f83ce680f8c0813cbb6654f037545f939c29288b0b0f222fe2a5

EXTENDED KEY:
a6d6dbdf4922f83ce680f8c0813cbb66
54f037545f939c29288b0b0f222fe2a5
b24edd4cfb6c25701decddb09cd066d6
8a8004a2d513988bfd989384dfb77121
19ed20d2e28105a2ff6dd81263bdbec4
71faaabea4e932355971a1b186c6d090
a99d40964b1c4534b4719d26d7cc23e2
7fb18c26db58be1382291fa204efcf32
7e176364350b2650817abb7656b69894
ceffca0415a77417978e6bb59361a487
815e74b8b45552e8352fe99e6399710a
3511696320b61d74b73876c12459d246
6aeb2e8edebe7c66eb9195f88808e4f2
f12100ead1971d9e66af6b5f42f6b919
68bdfaa2b60386c45d92133cd59af7ce

CONSTRAINTS ON ROWS:
00000000000000000000000000000000000000000000000000000000000000000
00000000000000000000000000000000000000000000000000000000000000000
00000000000000000000000000000000000000000000000000000000000000000
00000000000000000000000000000000000000000000000000000000000000000
00000000000000000000000000000000000000000000000000000000000000000
00000000000000000000000000000000000000000000000000000000000000000
00000000000000000000000000000000
725ae620310aa626f80a6e83ad143ec9d7da6ca8663721c8db2fbbe329f06135
1ba555aa73a0adfc01e484755293d72c339cbe51d73b3d0a1cf3c9144ac74ce0
cfc9025c88a86212a965485553775359b9c9446b0e699259bd2faf40563485f4
37e643dddf7508d58ada509bfa121b0c2c8f0ce07ee86e6210b8e7c3eb1b2ab4
70261bbd0a7394cc469283ac70c84b97e70be3d386acd31dc305c0e2fba3cd77
751b4f74734dce671dfa2efa365ac83b1ec391698458da55e6f5661e38a60d95
59d51baf8224a0911ec625962ba0e6c1d96954ba4119ddd6fcf2716676573a76
```

Weitere Informationen:
http://mweissbacher.com/blog/tag/truecrypt/

Stuxnet Internas:

Ein Windows XP nutzt lediglich eine Instanz von lsass.exe

pstree

Dienst	PID	PPID	Datum	Uhrzeit
Lsass.exe				
Lsass.exe				
Lsass.exe				
Services.exe				
Winlogon.exe				

Der echte lsass Prozess hat die höchste Priorität

threads

Dienst	PID	TID	Base Priority
Lsass.exe			
Lsass.exe			
Lsass.exe			

Der echte lsass Prozess lädt die meisten DLL's

dlllist -p 680,868,1928

Dienst	PID	Anzahl DLL's
Lsass.exe		
Lsass.exe		
Lsass.exe		

Nicht originale DLL's (also von Drittanbieter) erscheinen nicht in Services.exe, Lsass.exe oder Explorer.exe
Daraus folgt, dass der originale lsass keinen exe Header besitzt und keine MM_EXECUTE_READWRITE Berechtigungen hat.

malfind -p 680,868,1928

Stuxnet ruft über LoadLibrary Dateien auf, die NICHT auf der Festplatte existieren und über VirtualAlloc oder WriteProcessMemory geladen werden.

Die gefunden Einträge weisen beim ldrmodules Befehl KEINE Pfade auf sondern sind leer !

```
ldrmodules -p
```

Dienst	PID	Basisadresse
Lsass.exe		
Lsass.exe		
Lsass.exe		

Stringvergleich originaler lsass und die beiden anderen:

```
procexedump -p 680,868,1928 -D
strings
```

Stuxnet legt folgende Dateien an: oem7a.pnf mdmeric3.pnf mdmcpq3.pnf

```
filescan
```

Stuxnet nutzt zwei Treiber: mrxnet.sys und mrxcls.sys

```
modules
```

Im Originalcode kommt der String b:\myrtus\src..... vor

Dieser String muss im Kernel Memory gesucht werden!

```
yarascan -K -Y "myrtus"
```

Memory Dump mit winpmem:

Ein Memory Dump mit winpmem (winpmem_v3.3_rc.3.exe) erzeugt standardmäßig ein komprimiertes AFF4 Format. Soll der Dump mit Volatility ausgewertet werden, so muss ein raw Format erstellt werden:

```
winpmem.exe --mode PhysicalMemory --format raw --volume_format
raw -m -o ram.dmp
```